not interacting or becoming entangled with anything. So as a generous overestimate, let's imagine that every particle in the universe interacts and splits the wave function in two a million times per second, and has been doing so since the Big Bang, which was about $10^{18}$ seconds ago. That's $10^{88} \times 10^{6} \times 10^{18} = 10^{112}$ splittings, producing a total number of branches of $2^{10^{112}}$.

"Nice!" Alice seemed pleased with herself. "That's still a really big number, but it's much smaller than the number of dimensions in the Hilbert space of the universe. Pitifully smaller, really. And it should be a safe overestimate of the number of branches required. So even if the question of how many branches there are doesn't have a definite answer, we don't need to worry that Hilbert space is going to run out of room."

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"Well, good, I was worried there for a second." Her father's martini tasted pleasantly briny from the olives. He regarded Alice, a glint in his eye. "Had you really never asked yourself that question before?"

"I think most Everettians train themselves to think of the relative weights of various different branches of the wave function, rather than actually counting anything. We don't know the ultimate answer, so it doesn't seem too fruitful to worry about it."

"I'll have to process this a bit, because I always thought that there were supposed to be an infinite number of worlds, and that Many-Worlds implied that everything happened somewhere. That every possible world exists out there in the wave function. I thought that was the selling point. When I was stuck on a calculation, it was comforting to think that there was another world in which I was a llama, or a genius billionaire playboy philanthropist."

"Wait, you're not?" Alice feigned surprise. "I always thought you looked a bit like a llama."

"I mean, for that matter, in some world I should be a billionaire llama."

"Before we get off track," she continued, "let me just note that it's not 'you' who would be a llama or a billionaire, those would be other beings entirely. I'm sure we'll come back to that. But of more direct relevance to the issue, Many-Worlds doesn't say 'everything possible happens'; it says